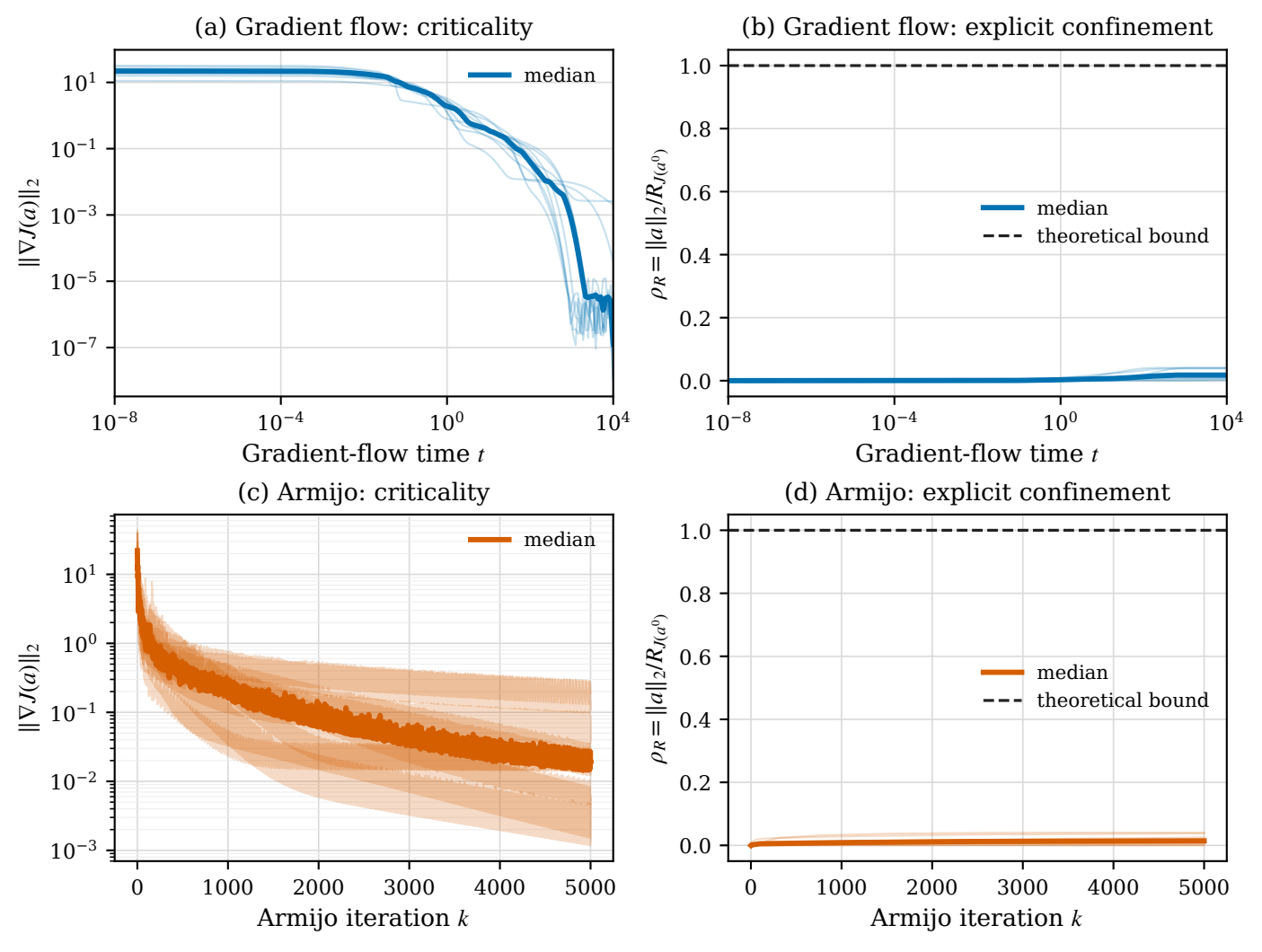

(a) Gradient flow: criticality
median
$\|\nabla J(a)\|_2$
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-8}
10^{-4}
10^{0}
10^{4}
Gradient-flow time t
(b) Gradient flow: explicit confinement
median
theoretical bound
$\rho_R = \|a\|_2 / R_{J(a^0)}$
1.0
0.8
0.6
0.4
0.2
0.0
10^{-8}
10^{-4}
10^{0}
10^{4}
Gradient-flow time t
(c) Armijo: criticality
median
$\|\nabla J(a)\|_2$
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
1000
2000
3000
4000
5000
Armijo iteration k
(d) Armijo: explicit confinement
median
theoretical bound
$\rho_R = \|a\|_2 / R_{J(a^0)}$
1.0
0.8
0.6
0.4
0.2
0.0
0
1000
2000
3000
4000
5000
Armijo iteration k